gbareader 10.0

Instructions

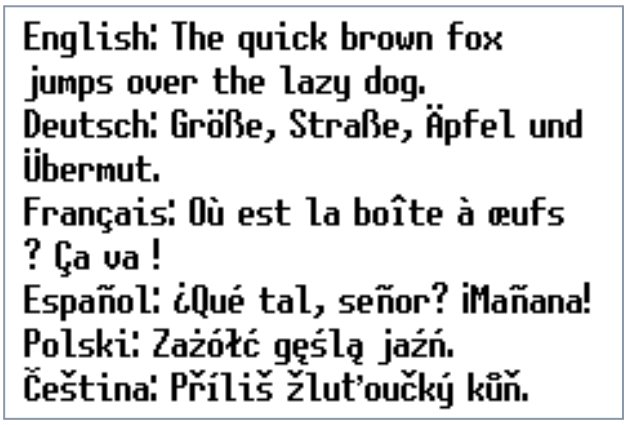
gbareader reads books (**.txt** and **.epub**) and comics (**.gbcomic**) on a Game Boy Advance with a SuperCard SD, EZ-Flash Omega or EverDrive GBA. It remembers where you stopped, shows text in many alphabets and lets you turn the screen for holding the console sideways.

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1. Getting started
2. Reading books
3. Settings
4. Comics: converting on a computer
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6. Supported alphabets and languages
7. Tips and problems
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Home: the books in /gbareader



A book open

1. Getting started

What you need

• A Game Boy Advance, GBA SP or another console that runs GBA games, with a **SuperCard SD**, **EZ-Flash Omega** (start the ROM from the SD card, not from NOR) or **EverDrive GBA**, and an SD card.

• The file **gbareader.gba**, free at **halimj.itch.io** (copy it anywhere on the SD card and start it from your flashcart's menu).

• Books: UTF-8 text files (**.txt**) and EPUB e-books (**.epub**, text only, without DRM).

• Comics: **.gbcomic** files made with the gbcomic conversion software on a computer (**halimj.itch.io**, chapter 4).

Putting books on the card

1. Make a folder called **gbareader** in the top level of the SD card.
2. Copy your **.txt**, **.epub** and **.gbcomic** files straight into it (not into subfolders).
3. Start gbareader. The books appear on Home, sorted by name, eight at a time.

Books elsewhere on the card can be copied in from the GBA: press **Start** on Home to open **Import**, find the book, press **A** and choose **Yes**.

Home and Import

Button	Home	Import
Up / **Down**	Choose a book	Choose a folder or book
Left / **Right**	Eight books back / on	
A	Open the book	Open the folder, or ask **Import into /gbareader?**
B		One folder up (at the top: back Home)
Start	Import	
Select	Settings	
L+R	Sleep: screen off, almost no power; **Start** held 1 s wakes it	

Sleep (**L+R** on Home): the screen goes off and the GBA waits for **Start** held for 1 second, which only wakes it and shows the same list again; a shorter press leaves it asleep. Switch off as usual when you are done; sleep is for short breaks.

gbareader never changes your books. What it saves (your place, the settings of that book) goes into a companion file next to the book: **Title.txt.sav** or **Title.epub.sav**. Keep it with the book when you move or rename it.

2. Reading books

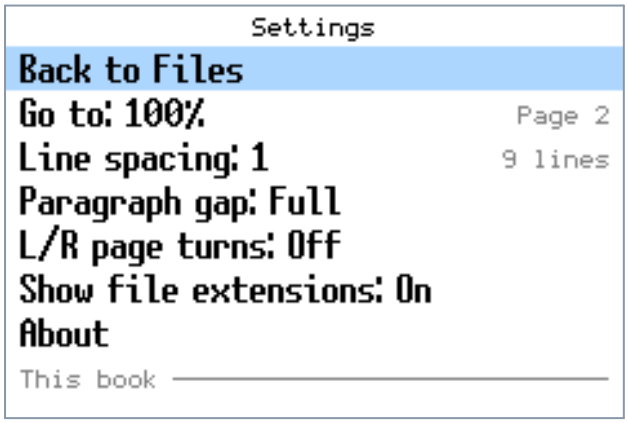
Button	What it does
A or **Right**	Next page
B or **Left**	Previous page

Button	What it does
R / **L**	Next / previous page when **L/R page turns** is On (Settings)
Hold **Up**	About 0.8 s: switch L/R page turns on or off (**L+R: On** / **L+R: Off**)
Start	Save your place (**Saved**). Do this before you switch off.
Select	Settings, with **Back to Files** at the top to close the book

With the screen turned (Settings > Angle), the D-pad turns with it: **Right** is whatever points right on the turned page. If you leave a book on a page you did not save, **Back to Files** first asks **Continue without saving?**: **A** leaves, **B** stays.

3. Settings

Select opens Settings. **Up** / **Down** choose a row, **Left** / **Right** change it, **A** steps it on; **B**, **Select** or **Start** go back. The grey text at the right shows the page and the lines per page.



Settings while reading a book

Row	What it does
Back to Files	Close the book and go Home
Go to	Jump to a place: books by percent (**Left**/**Right** 1%, **L**/**R** 10%), comics by page number; **A** jumps
Line spacing	0 to 4 (all books)
Paragraph gap	None, Small, Half or Full (all books)
L/R page turns	Shoulder buttons turn pages too in books; in a comic it is called L/R zoom (held): holding **R** zooms in, **L** zooms out
Show file extensions	On: Home shows **.txt** / **.epub**; Off hides them
About	Who made it, the controls in short, the credits
This book: Angle	**A** opens the Angle screen: Normal, 90° right, 90° left, Upside down; **A** turns the screen at once. **Left**/**Right** on the row step through them. Each book keeps its own.
Page map	Comics only (shown while one is open): the small page map on or off, for all comics
Animations	Comics only (shown while one is open): how screens move, for all comics, see chapter 5

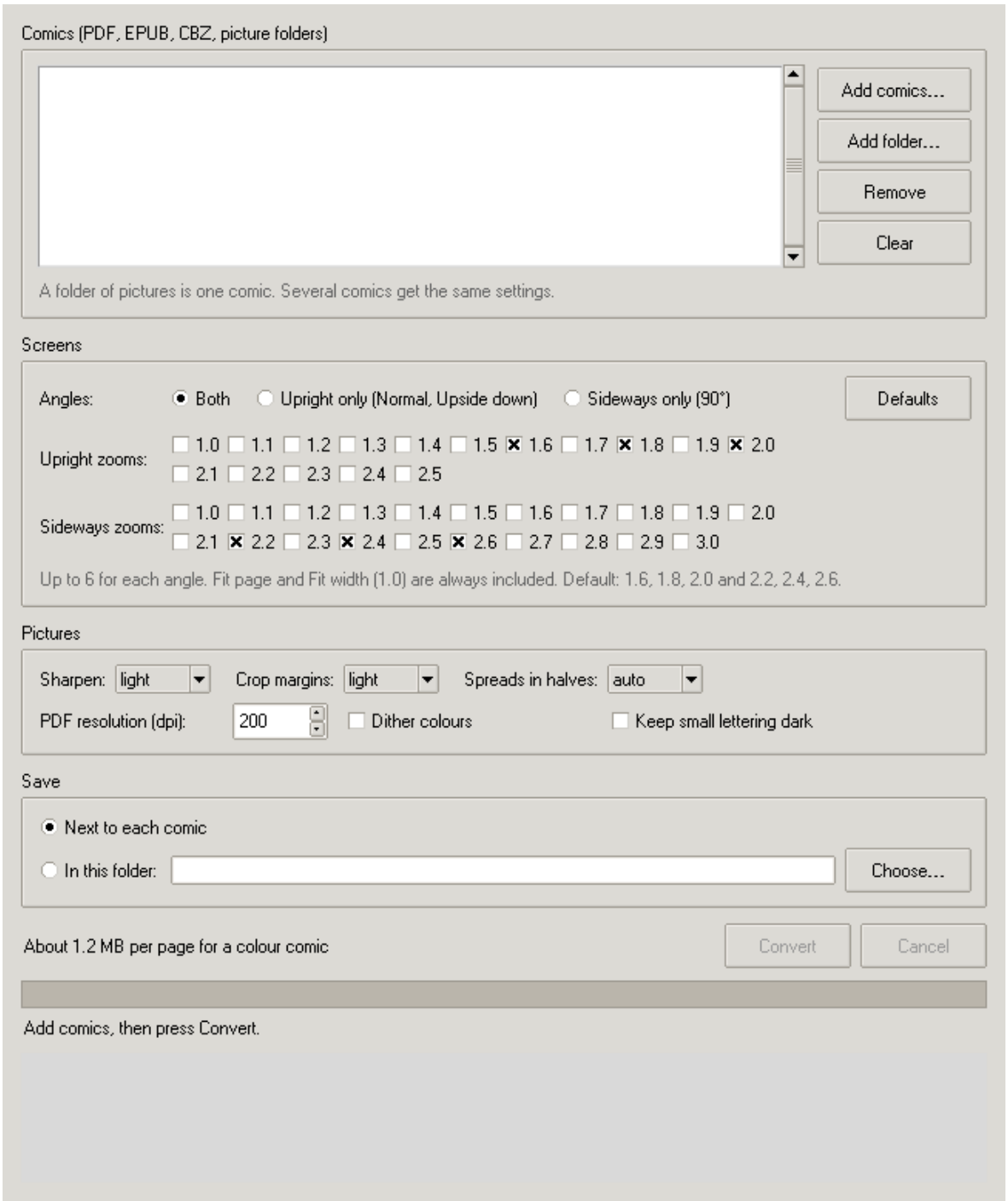
Row	What it does
This book: View	Comics only, see chapter 5

4. Comics: converting on a computer

The GBA cannot unpack PDF or CBZ pictures fast enough, so a comic is converted once on a computer into a **.gbcomic** file. The converter reads **PDF**, **EPUB** (comic EPUBs: their pictures), **CBZ** (or a ZIP of pictures), a **folder of pictures** or a single picture.

Starting the converter

- Conversion software: **halimj.itch.io**
- **Windows**: unpack **gbcomic-windows-x64.zip** and start **gbcomic.exe** (or drop comics on it). Nothing to install.
- **Linux**: unpack **gbcomic-linux-x86_64.zip** and start **gbcomic**.



The converter window

Converting

1. **Add comics...** (files) or **Add folder...** (a folder of pictures is one comic). Several comics get the same settings.

2. **Angles**: **Both**, **Upright only** (Normal and Upside down) or **Sideways only** (90°, the GBA held on its side). One kind of angle makes the file about half as big.

3. **Zooms**: tick any of 1.1 to 3.0 for each angle. Fit page and Fit width are always in the file. **Preview…** shows a page at a zoom on the GBA screen first (pages, A/B, D-pad, page map). Defaults: 1.6, 1.8, 2.0 upright and 2.2, 2.4, 2.6 sideways; **Defaults** brings them back. Bigger zooms make bigger files: the size is shown at the bottom.

4. **Pictures**: sharpening, cropping blank margins, wide pages in halves, PDF resolution, dithering and keeping small lettering dark. The defaults suit most comics.

5. **Save**: next to each comic, or in a folder you choose.

6. Press **Convert** and wait for the bar. Copy the **.gbcomic** into **/gbareader** on the SD card.

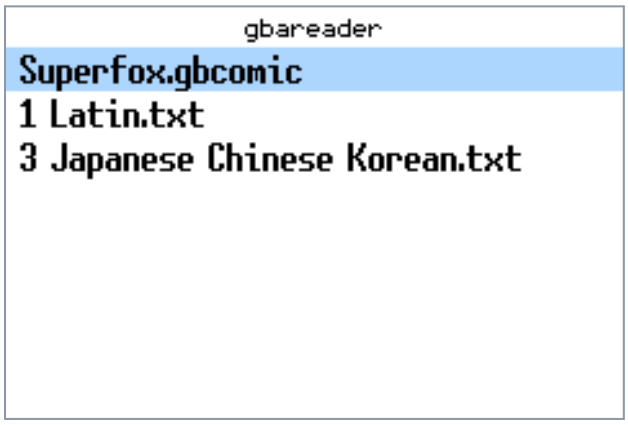
A colour comic with the defaults takes about 1.2 MB per page. A **.gbcomic** made by a test build before the final format must be converted again.

5. Comics: reading on the GBA

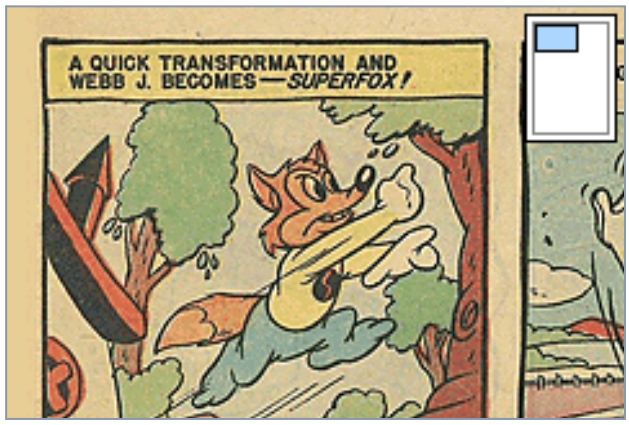
Each page is cut into a fixed grid of screens, the same on every page. **A** and **B** go from screen to screen and on to the next page; the D-pad moves freely around the picture.

Button	What it does
A	Next screen, after the last one the next page
B	Previous screen, before the first one the previous page
Hold **R** / **L**	When L/R page turns is On (it reads L/R zoom (held) in a comic): **R** the next bigger zoom, **L** the next smaller one, centred on the same place, while held; D-pad, **A** and **B** still work
D-pad	Move around the picture (held: keeps moving)
A after moving	The next screen of the same row, at the height you are at; at the end of the row the first screen of the next row
B after moving	The same way back; at the start of a row the last screen of the row above
Start	Save the page
Select	Settings

Right to left (manga) mirrors all of this; Columns reads down the columns. A D-pad still held when you press **A** or **B** does nothing until you let it go. Screens with only paper on them are passed over (**Skip empty**).



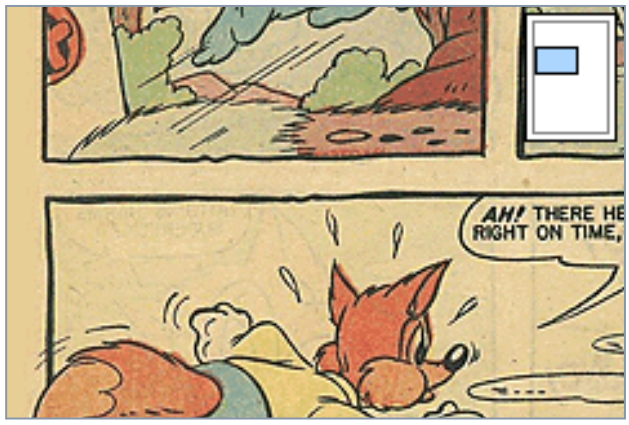
A **.gbcomic** on Home



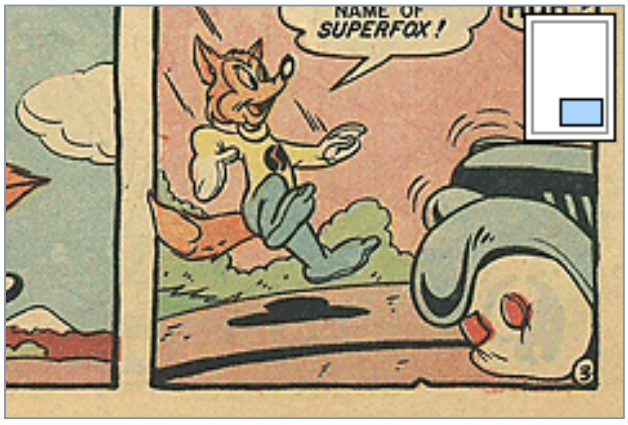
A opens it: the first screen (Zoom 1.6x)



A: the next screen of the row



A at the end of a row: the next row



D-pad: moved down; the page map shows where you are

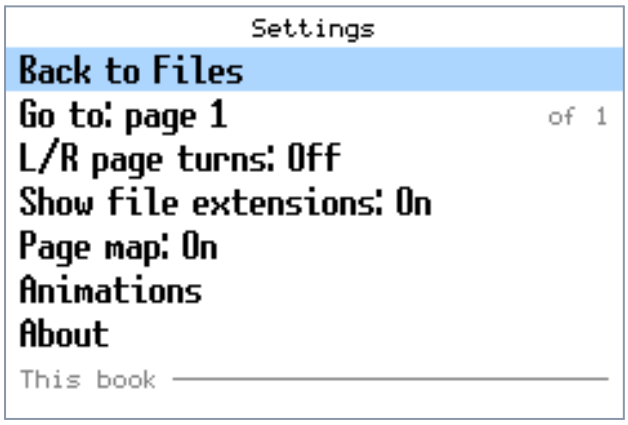


Angle 90° right: the GBA held on its side

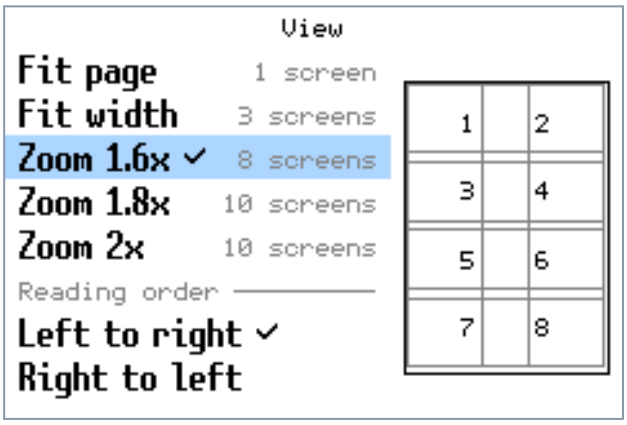
This book: View

Left / **Right** on the View row change the zoom. **A** opens the View screen: choose a zoom, the reading order (Left to right, Right to left, Columns), the number of **Columns** and **Rows** of screens (**Auto**, or 1 to 9) and **Skip empty**. The picture shows the page cut into numbered screens.

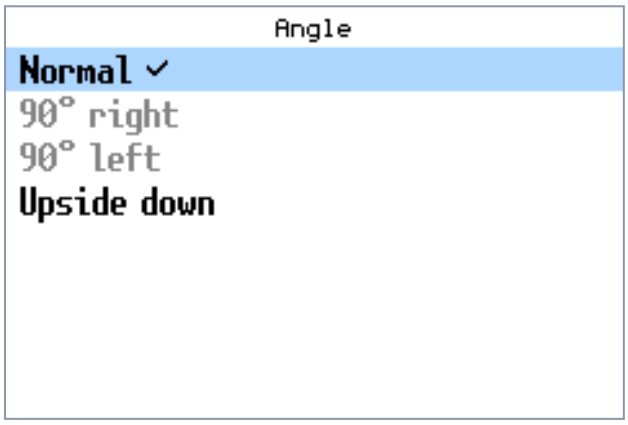
- Normal and Upside down share a zoom; 90° right and 90° left share another.
- Columns and Rows stay for every zoom and when you turn the screen; **Auto (n)** shows how many each zoom needs.
- Fewer screens than the page needs: the parts left over are shared evenly around the screens.



Settings of a comic: Page map and Animations with the general rows, View under This book



View: zooms with their screens; the page cut into 8 screens

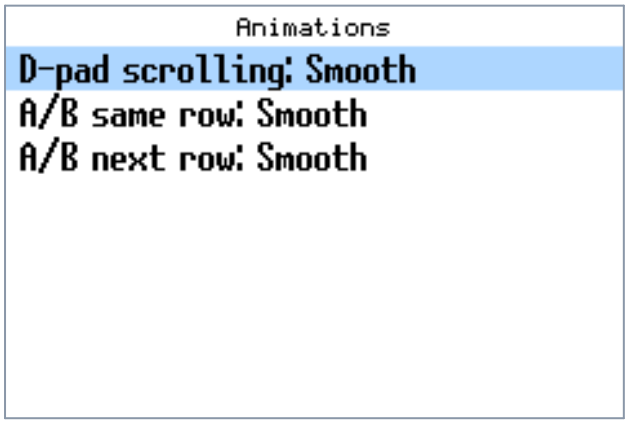


Angle: a comic made for upright screens only; the sideways angles are grey and skipped

Page map and Animations

• **Page map** (general setting, all comics) On: a small map at the top right shows where on the page you are, for a second after the screen settles.

• **Animations** (general setting, all comics): **D-pad scrolling**, **A/B same row** and **A/B next row**, each **Static** (jump), **Slide** or **Smooth** (speeds up and glides to a stop). All are Smooth to start with.



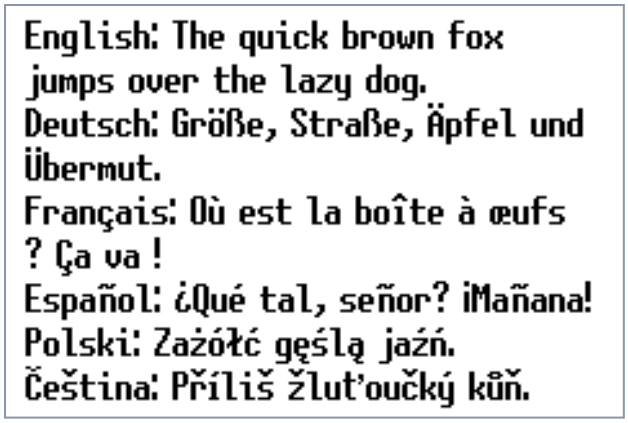
Animations

The sample comic is page 3 of a 1950s Superfox comic book.

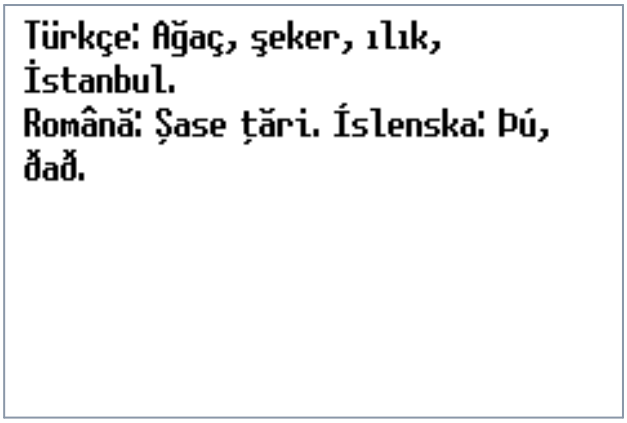
6. Supported alphabets and languages

Books are read as UTF-8. gbareader has pixel fonts for these alphabets (checked on the GBA with the sample books in **docs/guide/samples**):

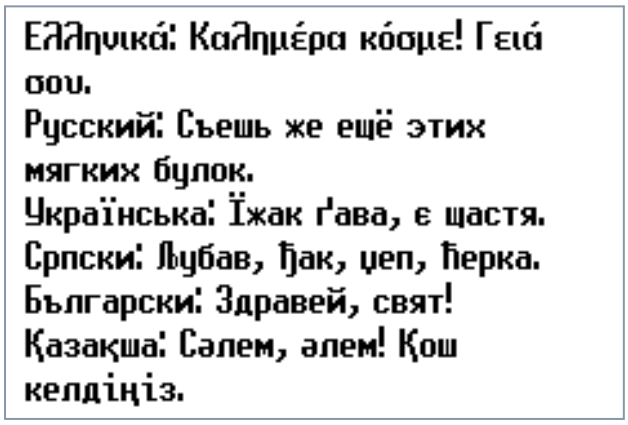
Alphabet	Languages, for example	What is covered
Latin	English, German, French, Spanish, Portuguese, Italian, Dutch, Nordic languages, Icelandic, Polish, Czech, Slovak, Hungarian, Croatian, Slovenian, Romanian, Turkish, Baltic languages, Albanian, Maltese, Esperanto, Latin	Basic Latin, Latin-1, Latin Extended-A and -B (U+0000–024F)
Greek	Modern Greek (monotonic)	Greek and Coptic (U+0370–03FF)
Cyrillic	Russian, Ukrainian, Belarusian, Bulgarian, Serbian, Macedonian, Kazakh, Kyrgyz, Tatar, Mongolian and more	Cyrillic (U+0400–04FF)
Japanese	Japanese	Hiragana, Katakana, about 21,000 kanji, Japanese comma and full stop
Chinese	Simplified and Traditional Chinese	CJK Unified Ideographs (U+4E00–9FEF) and the first 205 of Extension B
Korean	Korean	All 11,172 Hangul syllables
Arabic	Arabic	The 28 letters with hamza forms, joined (initial, middle, final), lam-alef ligatures, right-to-left lines; Arabic-Indic digits are shown as 0–9
Symbols		Punctuation, currency (€ £ ¥, rupee, rouble, won, lira ...), letterlike symbols (numero sign, ™ ...), fractions, degrees



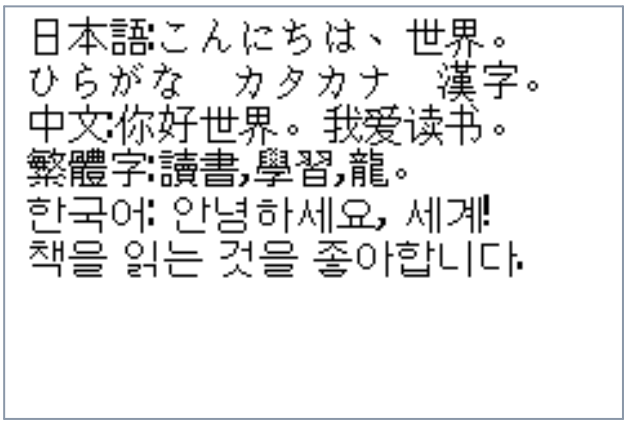
Latin: English, German, French, Spanish, Polish, Czech



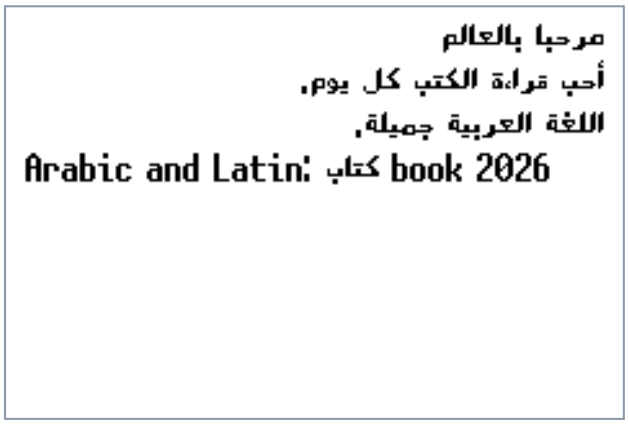
Latin: Turkish, Romanian, Icelandic



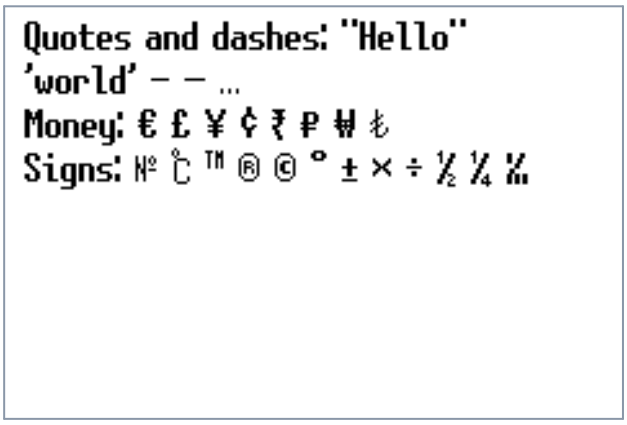
Greek and Cyrillic: Russian, Ukrainian, Serbian, Bulgarian, Kazakh



Japanese, Chinese (Simplified and Traditional), Korean



Arabic, joined and right-to-left, mixed with Latin



Quotes, dashes, currency and other signs

Fonts

All text in books uses 16-pixel-high pixel fonts. Which font draws which letters:

Font	Used for	By, license
SuperFW built-in font	Basic Latin: A–Z, a–z, digits, punctuation	David Guillen Fandos (SuperFW), GPL v3+
UNSCII 16 (full)	Accented Latin (Latin-1, Extended-A and -B), Greek, Cyrillic, punctuation, currency and letterlike symbols, Japanese kana, Chinese and Japanese ideographs (the ideographs come from GNU Unifont)	Viznut; GNU Unifont glyphs: Roman Czyborra, Paul Hardy and contributors; GPL
GNU Unifont Hangul	Korean: each syllable is put together from its letters	Unifont contributors, GPL v2+ with font exception
Ghoulam Regular	Arabic, 11 pixels high, with joined letter forms	Imad AlFil / mloukhiyye, CC BY 4.0
gbamp3 small font (5×7)	Menus, the Home list, messages: basic Latin only	Halim Jarrar (gbamp3), GPL v3+

There is no font setting: each letter takes the first font above that has it.

Several alphabets in one book

Any of these can be mixed on one page, even in one line. Arabic lines are aligned to the right; Latin words and numbers inside them stay left to right.

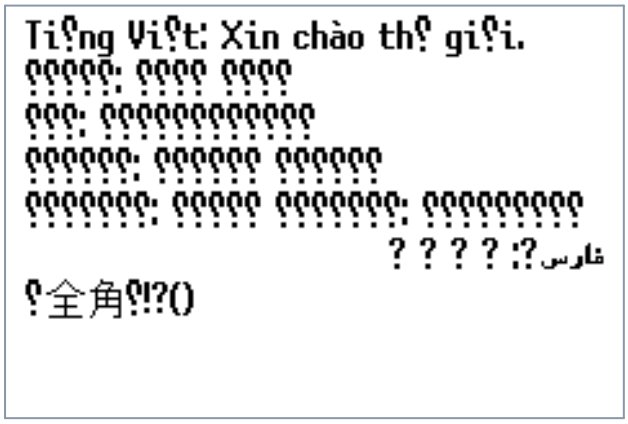
How Arabic is switched on

Arabic joining switches itself on when the page a book opens on has Arabic on it, and stays on for that book. If the Arabic only starts later in a book, go to an Arabic page, press **Start** to save, go **Back to Files** and open the book again. Vowel marks (harakat) are not shown; the text is not changed.

Not supported

Letters without a font show as **?**. Not supported are, among others:

- **Hebrew**, **Thai**, **Devanagari** (Hindi, Marathi, Nepali) and the other Indian scripts, **Armenian**, **Georgian**, **Ethiopic**.
- **Vietnamese**: most letters with tone marks (they are outside Latin Extended-B).
- **Persian, Urdu** and other languages that add letters to Arabic: the extra letters (for example peh, tcheh, gaf) show as **?**.
- Polytonic (ancient) Greek, the rarer CJK ideographs (Extension A, most of Extension B), Japanese corner brackets.
- Emoji.



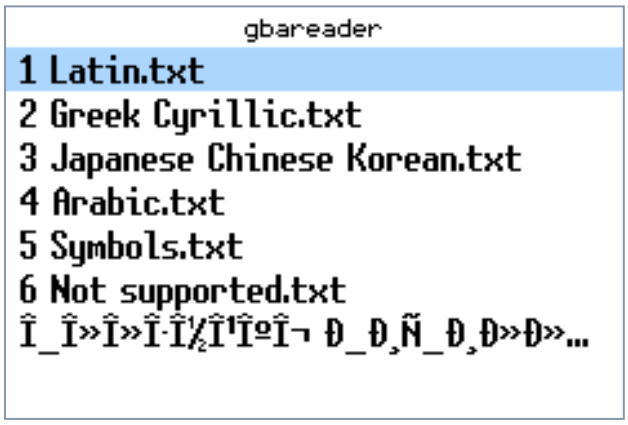
Vietnamese, Hebrew, Thai, Hindi, Armenian, Georgian, Persian extras: shown as ?

Small changes on screen

- Curly quotes become straight ones (" and '), long dashes become **-**.
- Full-width letters, digits and punctuation in Chinese and Japanese text (such as the full-width comma or colon) are shown as their ordinary forms.
- Only the screen changes; the book stays as it is.

File names

The Home list uses a small font with only the basic Latin letters A to Z, digits and punctuation. A book named in another alphabet (or with accents) still opens and shows correctly inside, but its name looks garbled on Home. Name your files with basic Latin letters.



The last book has a Greek, Cyrillic and Japanese file name

7. Tips and problems

You see	What to do
No books found	Make the folder **gbareader** at the top of the SD card and put the books directly in it; or press **Start** to import.
A book is missing on Home	Only **.txt**, **.epub** and **.gbcomic** directly in **/gbareader** are listed. CBZ and PDF must be converted to **.gbcomic**.
Position save failed, **Settings save failed**	The card is full, write-protected or the book name is too long (at most 251 bytes). Free some space or rename the book on a computer, then press **Start** again.
Continue without saving?	The page shown is not the one saved: **A** leaves anyway, **B** stays so you can press **Start** first.
? in the text	That alphabet has no font (chapter 6), or the file is not UTF-8. Save it as UTF-8 on a computer.
Arabic letters not joined	Open the book on an Arabic page and save (chapter 6).
A comic is too small or too big	Settings > View: another zoom; or convert again with other zooms.

Moving a book: move its **.sav** with it. Starting a book over: delete its **.sav** (the book stays as it is).

8. About / licence and credits

gbareader 10.0

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License: GPL-3.0-or-later. gbareader is free software: you can redistribute it and/or modify it under the terms of the GNU General Public License as published by the Free Software Foundation, either version 3 of the License, or (at your option) any later version.

No warranty. gbareader is distributed in the hope that it will be useful, but WITHOUT ANY WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE. See the GNU General Public License for more details.

Source code: github.com/Xinon232/gbareader

Credits and font licenses

Ghoulam Regular © 2025 Imad AlFil / mloukhiyye, CC BY 4.0. Converted actual contextual GSUB forms to native 11px monochrome ROM glyphs; this modified representation is not author endorsement. Source: https://mloukhiyye.itch.io/ghoulam-arabic-pixel-art-font-version-1 — License: https://creativecommons.org/licenses/by/4.0/.

SuperFW fonts, renderer and SD foundation: David Guillen Fandos, © 2024–2025, GPL v3 or later. https://superfw.davidgf.net/. The original SuperFW and supplemental font packs are unchanged.

UNSCII by Viznut: the included unscii-16-full-derived font pack is GPL, not wholly public domain. It incorporates GNU Unifont and public-domain Fixedsys Excelsior glyphs. https://viznut.fi/unscii/.

GNU Unifont / Hangul: Roman Czyborra, Paul Hardy and Unifont contributors. GPL v2 or later with the GNU font embedding exception. https://www.unifoundry.com/unifont/.

UI font: the 5x7 font from gbamp3 v0.9.8 by Halim Jarrar, part of this project family (GPL v3 or later). https://github.com/Xinon232/gbamp3.

SD card drivers (gbaSD, https://github.com/Xinon232/gbaSD): SuperCard SD driver from SuperFW by David Guillen Fandos, GPL v3 or later; EverDrive GBA driver ported from edio-gba by krikzz, MIT license; EZ-Flash Omega driver by Halim Jarrar, GPL v3 or later, written after the EZ-Flash Omega kernels (Apache 2.0).

Butano engine: Gustavo Valiente, zlib license. Built with devkitARM / devkitPro. FatFs © 2022 ChaN, permissive redistribution terms in gbasd/fatfs/ff.c. Framework dependency notices remain in the Butano source distribution.

miniz: MIT license; © 2010–2014 Rich Geldreich and Tenacious Software LLC; © 2013–2014 RAD Game Tools and Valve Software. Complete notice in third_party/miniz/LICENSE.

Full source and notices: https://github.com/Xinon232/gbareader. The app lists these credits on its About pages (Settings > About); Left/Right changes pages, B closes.